EDWARD LIU

Senior Fullstack Engineer (70% Frontend / 30% Backend) | Startup Visionary | Innovative Problem Solver

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TECHNICAL SUMMARY

- Diverse industry experience across AI, E-commerce, Healthcare, Real Estate, Financial, Retail, and Social.
- **Front-end development** expertise in **React**, **Next.js**, **Vue.js**, and mobile frameworks like **React Native** and **Expo**, implementing highly-organized state management with **Redux**, **MobX**, **Zustand**, and **Recoil**.
- Engineered **data visualization** solutions for dashboards using **Chart.js** and **D3.js**, enhancing data insights through improved UI/UX with **localization** and **web accessibility** principles.
- **Back-end development** with **Node.js**, **Express.js**, **Nest.js**, **Django**, **FastAPI**, and **Go**, utilizing both **monoliths** and **microservices** architecture; proficient in **database schema optimization** (**MySQL**, **PostgreSQL**, **MongoDB**) with **ORMs** (**Prisma**, **SQLAlchemy**).
- Implemented API layers with **RESTful APIs** and **GraphQL** and configured secure **authentication** using services like **Okta**, **AWS Cognito** and **Zitadel**; managed **cloud infrastructure** (**AWS**, **GCP**) with **IaC tools** (**Terraform**, **AWS-CDK**).
- Integrated third-party APIs (**Stripe API**, **HubSpot API**, etc.) into applications to enhance functionality.
- Committed to **Test-Driven Development (TDD)** using **Jest**, **Cypress**, **Pytest**, and **Playwright**, delivering high-quality code quickly while adapting to new technologies.
- Quick to deliver high-quality code and adapt to new technologies.

Key Skills:
HTML/CSS | Javascript | Typescript | React / Next.js | Node/Express | Python/FastAPI | Go | Postgres/MongoDB

EDUCATION

Master's in Computer Science

Columbia University in the City of New York

2014 - 2015 New York, NY

GPA **3.9** / 4.0

Bachelor's in Computer Science

Columbia University in the City of New York

2010 - 2014 New York, NY

EXPERIENCE

Senior Software Engineer (Contract)

Start Matter

01/2022 - Present Raleigh, NC

Led end-to-end development and successful delivery of large-scale, mission-critical projects for industry leaders in AI, Real Estate, Media, Retail, Finance, and other sectors.

- Developed high-performing, cross-platform frontend applications using **HTML/CSS**, **React.js**, **React Native**, **Next.js**, **TypeScript**, **TailwindCSS**, **Radix UI**, **Shadcn** and **MUI**.
- Enhanced UI/UX by customizing **Chart.js** and **D3.js**, integrating geo-components with **ArcGIS**, and improved the performance using **CDN** and using the appropriate combination of React server components in **Next App Router**.
- Implemented web accessiblity by following the **WCAG2** and multi-language support with **React-i18next**.
- Delivered efficient data fetching and caching strategies, leveraging **RESTful APIs** and **GraphQL** with **Javascript**, **Typescript**, **Python**, and **Go** based on a microservices architecture.
- Designed and utilized data models and query patterns for SQL and NoSQL databases, including **PostgreSQL**, **MongoDB**, **DynamoDB**, and **Redis**.
- Set up message brokers like **Kafka.js** and **RabbitMQ** for asynchronous communication between services, enhancing application responsiveness.
- Utilized **ElasticSearch** to provide personalized recommendation based on user's actions.
- Configured **CI/CD** pipelines with tools such as **Jenkins** and **CircleCI**, integrating them with GitHub for streamlined deployment processes.
- Implemented **Test-Driven Development (TDD)** using **Jest, Pytest** for unit testing and **Playwright** for end-to-end testing, while utilizing **BrowserStack** to ensure cross-browser compatibility.
- Built an application for documentation with **MKDocs** and customized **Material Theme**.
- **Led a team of eight engineers** across 7 different project initiatives.

EXPERIENCE

Senior Full Stack Engineer (Full-Time)

Noble AI

02/2019 - 01/2022 San Francisco, CA

Developed and integrated AI-driven clinical assistant capabilities into the platform, enhancing medical data analysis and decision-making processes for healthcare providers.

- Utilized **Vue.js, VueX** and **Vuetify** to create dynamic and interactive user interfaces, enabling healthcare providers to easily access and interpret AI-generated insights.
- Collaborated with AI specialists to implement AI-based clinical assistant functionalities within the platform, utilizing technologies such as **Python**, **FastAPI**, **TensorFlow**, and natural language processing algorithms.
- Designed and deployed **AWS Lambda** functions using a serverless framework with **Python** to facilitate the integration of machine learning models with electronic health records (**EHR**) systems.
- Developed and implemented **RESTful APIs** using **Go** to ensure seamless communication between frontend and backend systems and implemented documentation using **Swagger**.
- Integrated **PostgreSQL** database with **ORM** frameworks like **Prisma** to efficiently store and retrieve medical data for analysis and decision-making.
- Implemented DevOps practices using **Docker** and **Kubernetes** to streamline deployment processes and enhance the scalability and reliability of AI features on the platform.
- **Provided mentorship** to three junior to mid-level engineers.

Software Engineer (Full-Time)

Auth0

11/2016 - 02/2019 Bellevue, WA

Enhanced the Auth0 platform with improved UX/UI and security solutions, and developed a community platform.

- Developed over 60 standardized components using **React**, **TypeScript**, **Material UI**, and **Storybook**, and released them as a UI library for internal company use.
- Delivered a dynamic, user-friendly, high-performance multi-browser compatible frontend application utilizing **HTML/CSS**, **React**, **Next.js**, and **TypeScript**.
- Collaborated with design teams to translate pixel-perfect UI designs using **Figma**.
- Designed a database schema to support various roles and permissions, leveraging **PostgreSQL** and integrated it using **Prisma**.
- Set up **Apollo GraphQL** server and integrated **GraphQL** queries and mutations on the frontend.
- Maintained application with data visualization by connecting it to **DataDog.**
- Setup virtual environment and containerization using **Docker** and **Kubernetes**.
- Conducted unit testing using **Mocha** to ensure project quality.

Frontend Engineer (Contract)

Meta

04/2015 - 11/2016 New York, NY

Developed a gambling application that integrates multiple interactive games.

- Built lightweight web and mobile platform applications using **HTML/CSS**, **React**, **React Native**, **TypeScript**, and **Tailwind CSS**.
- Implemented detailed data visualizations on the dashboard and 20+ analytics pages using **D3.js**.
- Organized a well-structured state management workflow utilizing **Zustand**.
- Integrated over 50 backend APIs using **RESTful API** in **Laravel** backend and **Stripe API** for subscription services.
- Established **CI/CD** pipelines with **GitHub Actions** for streamlined deployment.
- Enhanced web accessibility and optimized browser performance to ensure user experience when handling large data volumes.
- Designed a highly maintainable and scalable architecture through well-structured UI, API services and database design.

STRENGTHS

Problem Guru

Approach challenges with startup grit and creativity to deliver solutions under tight deadlines.

Team Player

Galvanizes team success through mentoring junior developers and promoting collaboration.

SKILLS

FRONTEND

HTML/CSS | Javascript | Typesscript | React | Next.js | Vue | Nuxt.js | Remix | React Native | Expo | Angular | Redux | Recoil | MobX | Zustand | D3.js | Chart.js | MaterialUI | TailwindCSS | ShadCN | NextUI | Storybook | RadixUI | ChakraUI | KendoUI | AntD | Ionic | Figma | Sketch

BACKEND

Node.js | Express.js | Nest.js | Python | FastAPI | Django | Flask | Laravel | CI | Zend | Swagger | RESTful API | GraphQL | Go | Serverless | AWS SAM Template

DATABASE/ORM

MySQL | PostgreSQL | MongoDB | Dynamo | Redis | SQLite | Prisma | Loopback | SQLAlchemy

Cloud Services, Infrastructure & Testing

AWS | GCP | Docker | Kubernetes | ElasticSearch | Jest | Playwright | BrowserStack | GitHub Actions | Jenkins | CircleCI | Kafka | RabbitMQ | DataDog | Sentry